PHAM VAN MINH

FPGA & Digital IC Design Intern

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Hanoi, Vietnam

SUMMARY

Electronics and Telecommunications Engineering student at Hanoi University of Science and Technology (HUST) with strong interests in **FPGA design**, **embedded systems**, and **software-hardware** integration. Experienced in personal and freelance engineering projects spanning **Verilog/SystemVerilog, STM32, and web development**. Fast learner with strong problem-solving and practical engineering skills.

EDUCATION

Hanoi University of Science and Technology (HUST) — **Oct 2023 – Present**

Bachelor of Electronics and Telecommunications Engineering

- GPA: **3.44**/4.0 (Highest Semeter : 3.9/4.0)
- Average Conduct Score: 94/100 (Excellent)
- Awarded Type-A Academic Scholarship (Semester 2025.1)

ACADEMIC PROJECTS & COMPETITIONS

Custom High-Speed System Board (Cyclone IV GX FPGA & STM32H7) — **Mar 2026 - Apr 2026**

- Designed a multi-layer high-speed PCB integrating Cyclone IV GX FPGA and STM32H7 MCU with differential pair routing, length matching, and impedance-controlled traces. *(KiCad 9 | High-Speed PCB Design | PDN Analysis)*

High-Speed Camera Streaming via FPGA & Gigabit Ethernet — **Feb 2026 - May 2026**

- Built a real-time video acquisition pipeline on FPGA streaming camera frames to a host PC through Gigabit Ethernet using a custom UDP packet engine in hardware. *(Quartus | FPGA | GigE Vision | Verilog | 1Gbps Ethernet)*
- Implemented a custom SDRAM controller as a multi-frame buffer with clock domain crossing between camera, memory, and network interfaces.

Integrated FPGA-ESP32 IoT Control Hub with Web Dashboard — **Mar 2026 - Apr 2026**

- Developed a full-stack IoT system: FPGA ↔ ESP32 (SPI bridge) ↔ Firebase Cloud ↔ responsive web dashboard with real-time signal monitoring. *(Verilog | C++ | Firebase | WebSocket)*
- Wrote SPI Slave module in Verilog for bidirectional data exchange between ESP32 and FPGA peripherals.

STM32 Game Controller Console — **Mar 2026 - May 2026**

- Designed and built a handheld game console from scratch using STM32F103RCT6, 2.4" ILI9341 LCD driven via SPI with DMA for smooth 30+ FPS rendering.(*C/C++ | STM32F103 | ILI9341 LCD (SPI + DMA) | USB HID | KiCad*)

SOFTWARE DEVELOPMENT PROJECTS

EcoCollect — Circular Economy Mobile Platform — **May 2026**

- Led the mobile development for a startup team at Banking Academy of Vietnam, winning 1st Place (Grand Prize) at the Faculty-level Startup Competition. *(Flutter (Dart) | Firebase | Google Maps API | Firestore)*
- Implemented real-time radar map for nearby collectors, green-point reward system, voucher redemption, order history tracking, and user profile management with Firebase Authentication and Cloud Firestore backend.

Ocean Love — Interactive 3D Ocean Simulation — **2023 - 2025**

- Developed a premium interactive 3D ocean experience using Three.js with custom vertex/fragment shaders for realistic wave simulation and dynamic lighting.
- Implemented real-time Boids flocking algorithm for a flock of seagulls with separation, alignment, and cohesion behaviors around a detailed Black Pearl ship model.

SKILLS

- **Programming:** Python, C++, Verilog, SystemVerilog
- **Tools:** Quartus II, STM32CubeIDE, STM32CubeMX, Arduino IDE, KiCad, Proteus
- **Other:** Canva, LaTeX, Basic PCB Design, Embedded Systems Development
- **Languages:** English (TOEIC 535), Vietnamese (Native)

ACTIVITIES & LEADERSHIP

- Organization & Inspection Committee — HUST Youth Union (08/2024 – Present)
- Executive Committee — K68 First-Year Student Youth Union & Association (08/2023 – 06/2024)
- Volunteer & Extracurricular Activities (2023 – Present)